MILITARY SPECIFICATION

KNOBS, CONTROL (FOR USE WITH ELECTRONIC, COMMUNICATIONS, AND ALLIED EQUIPMENT)

This amendment forms a part of Military Specification MIL-K-3926B, dated 26 July 1958, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

2.1, SPECIFICATIONS, MILITARY: Add "MIL-F-18240 Fastener, Externally Threaded, 250 F, Self-locking Element For."

2.1, STANDARDS, FEDERAL: Add "FED-STD-H28/2 Screw Standard for Federal Services, Section 2, Unified Threaded Form and Thread Series for Bolts, Screws, Nuts, Tapped Holes and General Applications."

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3.1 Delete and substitute:

"3.1 Specification sheets and Military Standard (MS) sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets and MS sheets. In the event of any conflict between requirements of the specification and the applicable specification sheets or MS sheets, the latter shall govern."

Add the following paragraphs:

"3.4.1.2 Threads. All threads shall be right hand and in accordance with FED-STD-H28/2."

"3.4.2.1 Slip-clutch (insert). The outer portion of the slip-clutch insert shall be non-rotatable, metallic, and molded in the center of the knob body. The inner portion shall be a rotatable metallic insert incorporating the shaft hole in the center, and two UNC-3B threaded holes of a minimum of three threads. A 45 degree chamfer of 1/32 ± 1/64 inch shall be provided in the shaft hole."

3.4.3, line 2: After FF-S-200: Add "Self-locking setscrews shall pass the tests specified in 4.7.2.1."

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3.5, line 2: Delete "inclusive" and substitute "and 4.7.8 through 4.7.10."

3.5(h), after "(Any type of corrosion which will prevent the setscrews": Add ",spinner or slip-clutch mechanism,"

3.6.1, line 2: After "failure of the": Add "self-locking element,"

3.6.2: Delete "4.7.3.2.2 and substitute "4.7.3.1.1".

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ld the following paragraphs:

"3.6.8 Slip-clutch (insert). When knobs are tested as specified in 4.7.2.8 and 4.7.3.2.5, there shall be no movement of the outer portion of the insert within the molded knob, cracking of the knob or other failures. The slip-clutch shall function in the normal manner with no evidence of binding. The overriding torques shall meet the requirement indicated on the respective military standard.

"3.6.9 Threaded shaft-hole knobs. When threaded shaft-hole knobs with or without insert are tested as specified in 4.7.2.9 and 4.7.3.2.6, there shall be no movement of the insert (if applicable), cracking or other failures."

"3.12 Slip-clutch (vibration). When tested in accordance with the test specified in 4.7.8, there shall be no degradation of the slip-clutch knob. The overriding torque requirements as specified on the applicable MS sheets shall be met.

"3.13 Slip-clutch (life, rotational). When slip-clutch knobs are tested in accordance with 4.7.9, there shall be no degradation of the overriding torque and tolerance specified on the applicable MS standard.

"3.14 Slip-clutch (shock test). When tested in accordance with 4.7.10, there shall be no degradation of the slip-clutch knob. The overriding torque requirements as specified on the applicable MS standard shall be met."

3.10a and c: After "MS", Add "or specification sheet".

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TABLE I: Add new entry: "Self-locking element, 3.4.3, MIL-C-18240".

PAGE 5 AND 6

TABLE II: Delete and substitute:

TABLE II. Qualification inspection.

"

Examination or test	Requirement paragraph	Method paragraph
Group I (All sample units) Visual and mechanical examination	3.1.3.4 to 3.6 Incl 3.10 and 3.11	4.7.1

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TABLE II. Qualification inspection. (continued)

Group II (3 sample units from group I)		
Setscrews	3.6.1	4.7.2.1
Insert	3.6.2	4.7.2.2
Spinner and tactile spinner	3.6.3	4.7.2.3
Spinner-knob balance	3.6.4	4.7.2.4
Caps	3.6.5	4.7.2.5
Tactile caps	3.6.6	4.7.2.6
Dial skirts or plates	3.6.7	4.7.2.7
Slip-clutch (insert)	3.6.8	4.7.2.8
Threaded shaft-hole knobs	3.6.9	4.7.2.9
Group III (3 sample units from group I)		
Thermal shock	3.5	4.7.3
Moisture resistance	3.5	4.7.4
Finish	3.7	4.7.4.1
Vibration (slip-clutch)	3.12	4.7.8
Group IV (3 sample units from group I)		
Sunshine	3.5	4.5.5
Salt spray (corrosion)	3.6	4.7.6
Group V (3 sample units from group I)		
Flammability	3.8	4.7.7
Group VI (3 sample slip-clutch units from group I).		
Life (rotational)	3.13	4.7.9
Group VII (3 sample slip-clutch units from group I).		
Shock	3.14	4.7.10

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Table III, Subgroup II: Delete "(All lots-spinner types)" and substitute "(All lots-spinner and slip-clutch types)"; Delete "Insert" and substitute "Insert-normal". Add following line under "Insert", "Insert-slip-clutch, 3.6.8, 4.7.2.8".

PAGE 6 AND 7

Table III, Subgroup III: Delete "(All lots except spinner types and tactiles " and substitute "(All lots except spinner and slip-clutch types and tactiles)" in two places.

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Table III, Subgroup VI: As last line add "Threaded shaft-hole knobs, 3.6.9, 4.7.2.9"

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ble IV, Subgroup V: Add new entry "slip-clutch (insert), 3.6.8, 4.7.3.2.5".

ble IV, Subgroup VI: Add new entry "slip-clutch (insert), 3.6.8, 4.7.2.8".

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ible IV, Subgroup VII: Add new entry "Threaded shaft-hole knobs, 3.6.9, 4.7.2.9 and .7.3.2.6".

ible IV: Add the following:

Subgroup IX (3 sample slip-clutch units) Vibration	3.12	4.7.8
Subgroup X (3 sample slip-clutch units) Life (rotational)	3.13	4.7.9
Subgroup XI (3 sample slip-clutch units) Shock	3.14	4.7.10

4.7.2.1: Add the following as the last sentence "Self-locking screws shall be torque tested in accordance with the torque test requirements of MIL-F-18240."

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TABLE V: Delete Table V and substitute.

" TABLE V. Setscrew torque.

Setscrew Thread	Torque Reading in Inch Pounds
.086 - 56-UNC-3A	1.9
.112 - 40-UNC-3A	5.6
.164 - 32-UNC-3A	21

"

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Add the following paragraphs:

"4.7.2.8 Slip-clutch (insert). Slip-clutch knobs shall be mounted on a corrosion resistant shaft and held by a jig in a suitable manner to prevent rotation of the knob containing the non-rotatable portion of the insert. Prior to testing, the setscrews shall be seated to 1.5 inch-pounds torque for .086-56, 5 inch-pounds for .112-40 and 20 inch-pounds for .164-32. The slip-clutch shall function when a limit stop has been reached, the knob is in the detent position and rotation of the knob is continued in the same direction. The clutch mechanism shall override the stop without applying a torque to the shaft in excess of that specified on the applicable MS standard. The torque shall be applied in a clockwise and then in a counterclockwise direction for one minute in each direction.

"4.7.2.9 Threaded shaft-hole knobs. Knobs with threaded shaft-hole shall be secured on a threaded shaft to a torque of 15 inch-pounds minimum, and tested by applying a tensile force (axial pull) to the knob in accordance with the values in the table VI."

"4.7.3.2.5 Slip-clutch (insert). Slip-clutch knobs shall be tested as in 4.7.2.8."

"4.7.3.2.6 Threaded shaft-hole knobs. Plastic and plastic-insert threaded shaft-hole knobs shall be tested as in 4.7.2.9."

Table VI: Add to the title "or threaded shaft-hole knobs."

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"4.7.4.1, line 6: Delete "PP-T-60" and substitute "PPP-T-60."

Add the following paragraphs:

"4.7.8. Slip-clutch (vibration test). The slip-clutch knob, attached by its normal mounting means or equivalent, shall be subjected to vibration testing, Method 204.Condition A of MIL-STD-202.

"4.7.9 Slip-clutch (life, rotational). The slip-clutch knob, held in a suitable jig and in normal operating position, shall be subjected to Test Condition D, Method 206 of MIL-STD-202. Overriding torque shall be taken at ambient room temperature, prior to and after life rotational test. The knob shall be cycled in no-load condition at the rate of 20 cycles/minute for a total of 10,000 cycles. (see 3.13).

"4.7.10 Slip-clutch (shock test). The slip-clutch knob, held as normally mounted in service or equivalent, shall be subjected to shock testing Method 213, Condition G of MIL-STD-202."

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Add the following definition:

"(x) Slip-clutch -An integral part of the knob which functions by overriding, when the knob core part reaches the turning limit of the device to which it is attached."

Custodians:
Army - ER
Navy - EC
Air Force - 11

Review activities:
Army - AS, AT, AV, EA, ME, MI
Navy - MC, SH
DLA - IS

User activities:
Navy - CG, OS

Preparing activity:
Navy - EC

Agent:
DLA - IS

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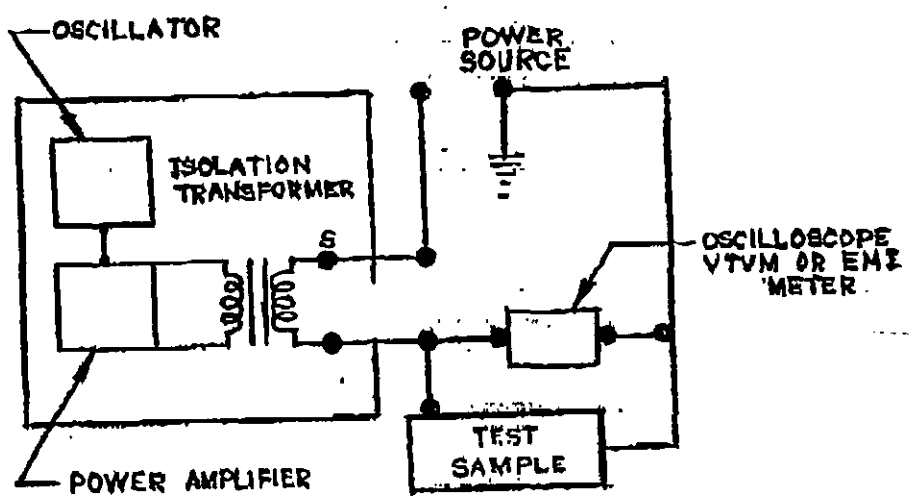


FIGURE CS01-1. Conducted susceptibility, 30 Hz to 50 kHz typical test setup

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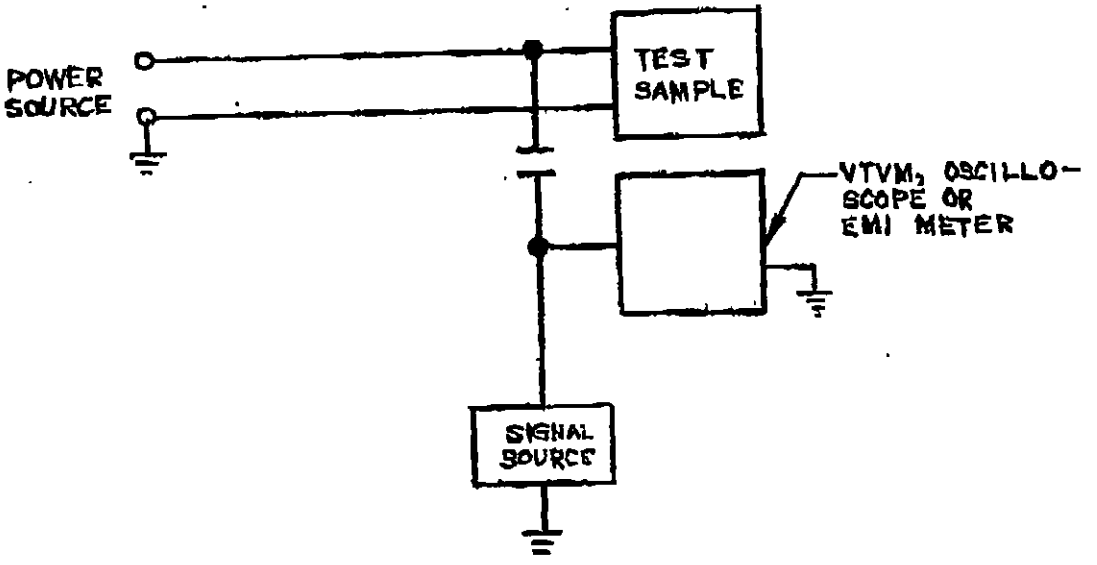


FIGURE CS02-3. Conducted susceptibility 50 kHz to 400 MHz

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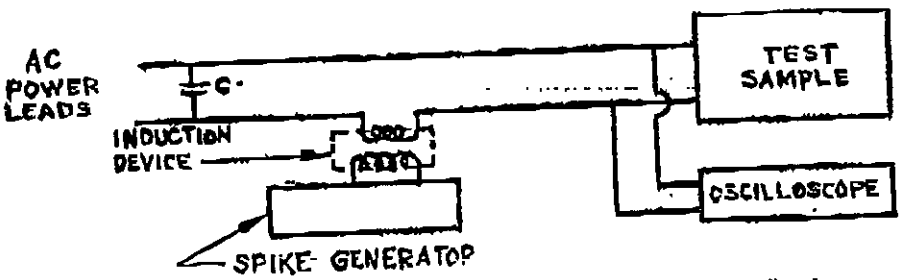


FIGURE CS06-1. Conducted susceptibility, spike, power leads; series injection

L = 20 μh

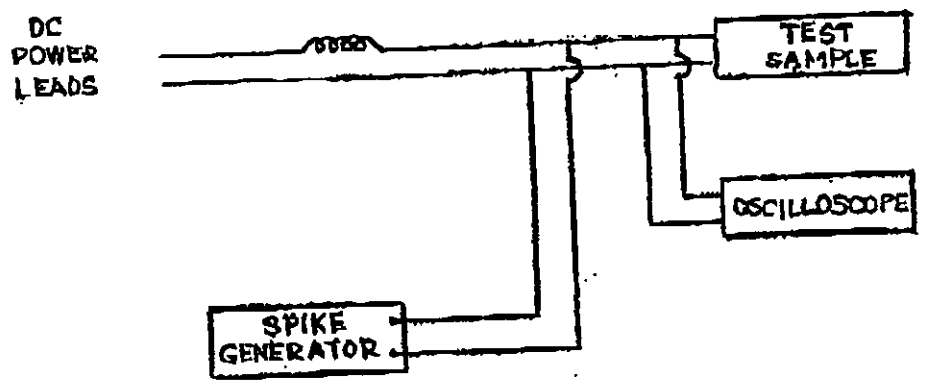


FIGURE CS06-2. Conducted susceptibility, spike, power leads; parallel injection

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RADIATED EMISSIONS

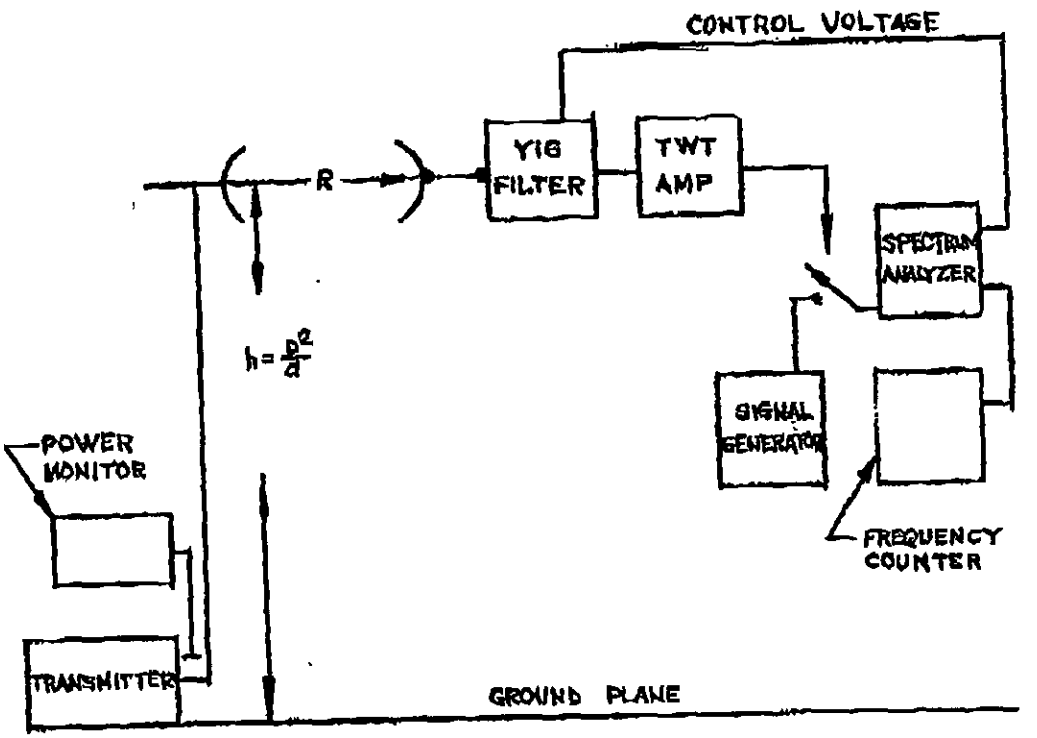


FIGURE RE03-1. Typical test setup -- spurious and harmonic emissions

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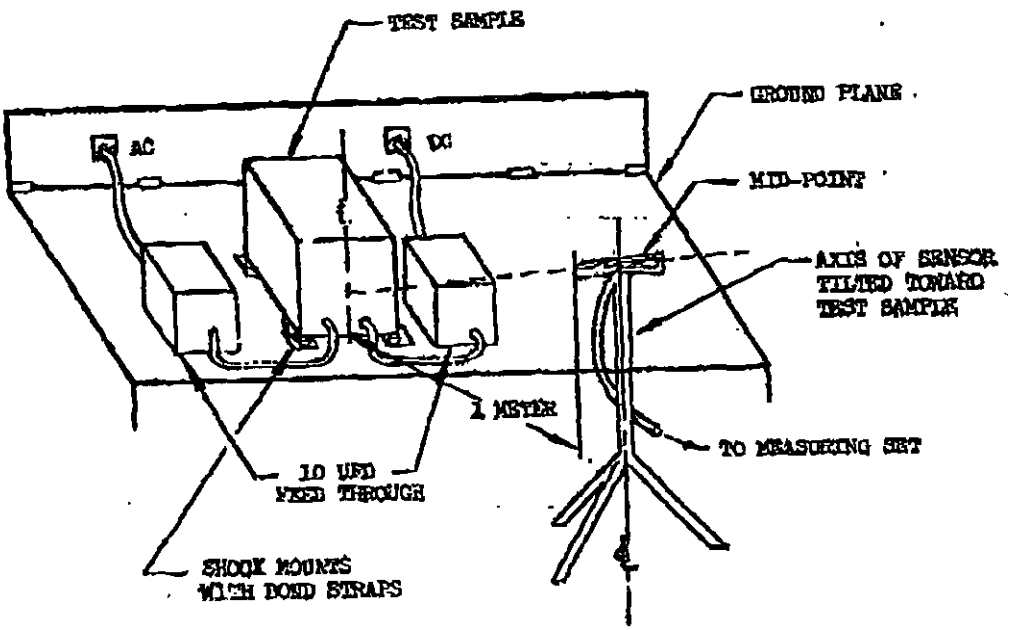


FIGURE (7) RE04-2. Typical test setup for radiation measurements (magnetic field sensor)

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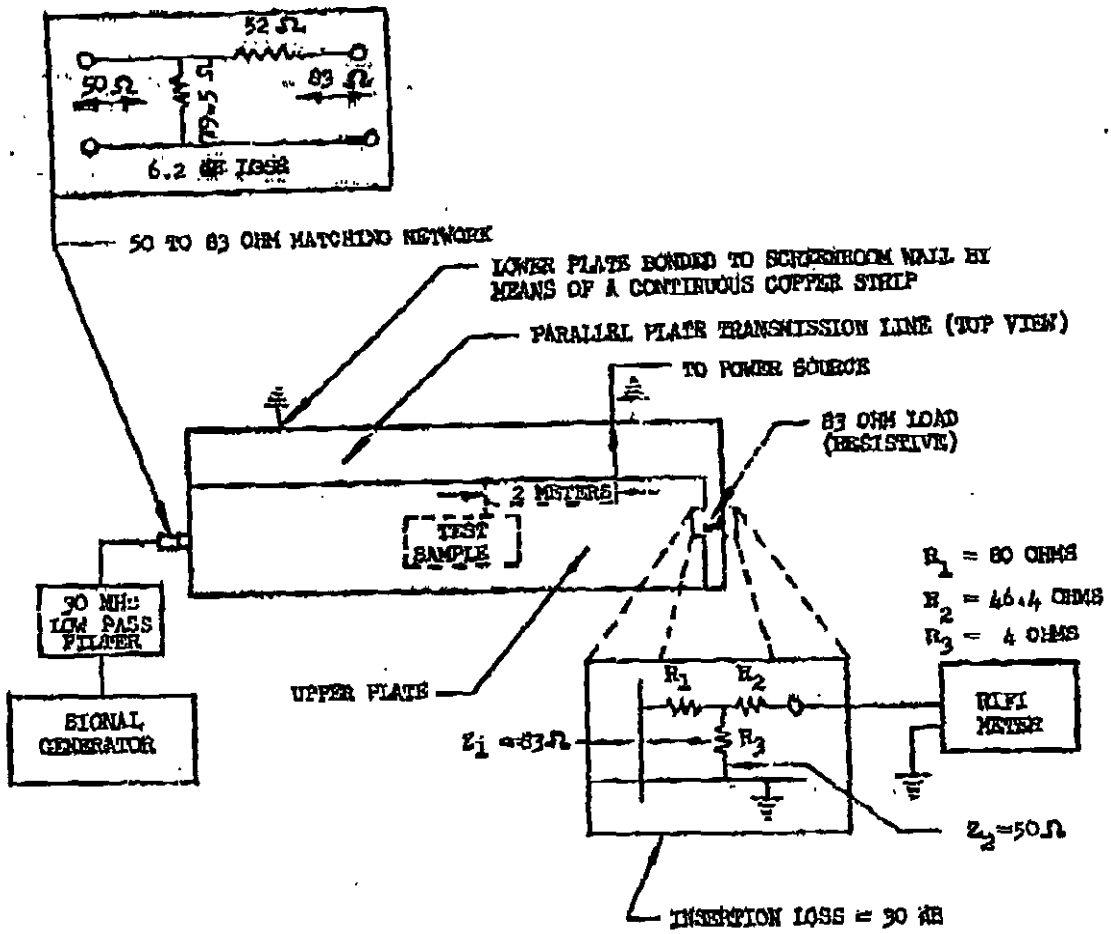


FIGURE RS04-3. Typical setup for susceptibility tests using parallel plate line

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